DriveSafe: A Computer Vision-Based Driver Drowsiness and Road Monitoring System

Prof. Trupti Shinde, Arjun Deokhile, Shreyash Nikam, Saish Patil, Krishna Soni, Neel Sharma
Department of Computer Engineering
Vishwakarma University
Pune, India

***Abstract*—Driver fatigue and reduced attention are important contributors to road-safety risk. This paper presents DriveSafe, a computer vision-based driver and road monitoring system designed to identify visual signs of drowsiness and support situational awareness using camera input. The driver-monitoring pipeline uses facial landmarks and the Eye Aspect Ratio (EAR) to estimate eye closure over consecutive video frames. MediaPipe is used for facial landmark processing, while OpenCV handles video capture and image processing. The broader system design supports a dual-camera arrangement: an inward-facing camera monitors the driver, and a forward-facing dash camera can be used for road-scene analysis, including object and lane detection. YOLOv8 and PyTorch are included in the road-monitoring technology stack, with a Next.js and Tailwind CSS interface for presenting system status and warnings. In the reported evaluation of the driver-state classifier, the system achieved 92.4% accuracy, 91.8% precision, 93.1% recall, and 92.4% F1-score on more than 2,000 test frames. These results describe the evaluated driver-state classification task and should not be interpreted as a validated measure of crash reduction or complete road-scene performance.**

***Index Terms*—Driver drowsiness detection, computer vision, Eye Aspect Ratio, facial landmarks, MediaPipe, OpenCV, YOLOv8, road monitoring, lane detection.**

I. Introduction

Driver drowsiness can reduce attention, reaction time, and the ability to respond to changing traffic conditions. A camera-based monitoring system offers a non-contact way to observe visual indicators of driver fatigue. However, a practical system must account for changes in illumination, head pose, facial movement, camera placement, and temporary occlusion.

DriveSafe is a computer vision-based prototype intended to monitor the driver and provide a visible warning when the observed eye state indicates possible drowsiness. The project also considers a dual-camera configuration to separate driver-facing analysis from forward-facing road analysis. The driver-facing pipeline focuses on facial landmarks and eye closure, while the forward-facing pipeline can analyse road objects and lane markings.

The main contributions of this project are:

- A real-time driver-state monitoring pipeline using facial landmarks and the Eye Aspect Ratio.
- A consecutive-frame decision rule to reduce alerts caused by brief eye closures or noisy detections.
- A dual-camera system design that separates driver monitoring from forward road-scene monitoring.
- An application interface intended to display driver state and prominent warnings.
- An evaluation summary for the driver-state classification task using accuracy, precision, recall, and F1-score.

II. Related Work

Early vision-based face detection approaches, including the Viola–Jones framework, established efficient methods for locating faces in images [1]. Facial landmark methods subsequently enabled more detailed analysis of eye and mouth geometry. Soukupová and Čech described an Eye Aspect Ratio-based method for detecting eye blinks from facial landmarks [2]. The EAR is computationally inexpensive and can be used as a component of drowsiness monitoring, although a low EAR alone does not prove that a driver is fatigued.

MediaPipe provides real-time perception pipelines and face landmark capabilities suitable for interactive applications [3]. For forward-facing road analysis, single-stage object detectors such as the YOLO family provide a practical basis for detecting vehicles, pedestrians, and other road users [4]. Lane detection is a separate perception task and may require a dedicated lane-marking model or image-processing pipeline. DriveSafe combines these ideas at the system-design level while keeping driver-state and road-scene evaluation distinct.

III. System Architecture

The proposed system is organised into four logical components:

1) **Driver-facing camera:** Captures the driver's face and eye region.
2) **Driver-state processing:** Uses OpenCV and facial landmarks to calculate EAR and classify the observed state as Alert or Drowsy.
3) **Forward-facing camera:** Provides road-scene frames for object detection and lane analysis. YOLOv8 is part of the technology stack for object detection; lane detection should be evaluated separately from driver-state classification.
4) **User interface:** A Next.js and Tailwind CSS frontend presents the monitoring state and warning information.

The dual-camera setup is intended to keep driver monitoring independent from road-scene monitoring. Each camera stream should be processed and evaluated separately, since the two

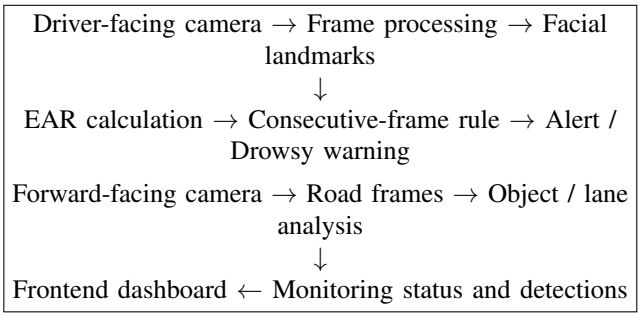


Fig. 1. High-level DriveSafe system workflow. Replace this conceptual block diagram with a measured implementation diagram if one is available.

streams have different inputs, labels, and performance requirements.

IV. Methodology

A. Video Capture and Preprocessing

Video frames are acquired from the driver-facing camera. OpenCV is used for frame capture and image operations. The implementation should verify that the selected camera is available, handle dropped frames, and keep the frame-processing loop responsive. Image resizing and colour conversion may be applied as required by the landmark model.

Lighting changes, motion blur, glasses, head rotation, and partial face occlusion can reduce landmark reliability. For robust evaluation, test data should include representative conditions and should not contain near-duplicate frames split across training and test sets when such duplication could inflate results.

B. Facial Landmark Detection

MediaPipe facial landmark processing estimates points around the eyes. Let the six landmarks associated with one eye be denoted by $p_1, \ldots, p_6$, arranged so that p_1 and p_4 are the horizontal eye corners and the remaining points describe the upper and lower eyelid positions.

C. Eye Aspect Ratio

The Eye Aspect Ratio measures the vertical opening of the eye relative to its horizontal width. For six eye landmarks, it is defined as

$$EAR = \frac{\|p_2 - p_6\| + \|p_3 - p_5\|}{2\|p_1 - p_4\|}. \tag{1}$$

When the eye is open, the EAR is generally higher; as the eyelids close, the value decreases. The threshold is dependent on the landmark convention, camera geometry, and individual user, so it should be calibrated and validated rather than treated as a universal constant.

D. Temporal Decision Rule

A single low EAR measurement may result from a blink or a noisy landmark estimate. DriveSafe therefore uses a threshold and a consecutive-frame condition. Let E_t be the measured EAR at time t, τ the selected threshold, and K the required number of consecutive low-EAR frames. A simplified decision rule is

$$D_t = \begin{cases} 1, & E_t < \tau \text{ for at least } K \text{ consecutive frames,} \\ 0, & \text{otherwise,} \end{cases} \tag{2}$$

where $D_t = 1$ indicates a drowsiness warning state. The threshold τ and frame count K must be chosen using validation data and the intended frame rate. The state should be reset when reliable open-eye observations resume. This rule is a heuristic for detecting prolonged eye closure; it is not a medical diagnosis of fatigue.

TABLE I
DriveSafe Technology Stack

Component	Technology / role
Programming	Python
Image processing	OpenCV
Facial landmarks	MediaPipe
Driver-state logic	Eye Aspect Ratio and temporal thresholding
Road object detection	YOLOv8, PyTorch
Frontend	Next.js, Tailwind CSS
Camera inputs	Driver-facing camera and forward-facing dash camera

E. Forward Road-Scene Analysis

The forward-facing camera is intended to provide road-scene context. YOLOv8 can be used to detect objects such as vehicles and pedestrians, while lane detection identifies lane boundaries or markings. Object detection and lane detection are different tasks: an object detector does not automatically provide reliable lane geometry. Each module should therefore use task-appropriate annotations and metrics, such as precision/recall or mean average precision for object detection and lane-specific accuracy or intersection-over-union measures for lane segmentation.

The driver-state metrics reported in Section VI apply only to the evaluated driver-state classification task. No separate quantitative result is claimed here for road-object detection or lane detection unless those modules are independently tested.

F. Interface and Warning Presentation

The frontend is built with Next.js and Tailwind CSS. Its role is to present monitoring status and warnings clearly. A drowsiness warning should be visually prominent, while normal operation should be distinguishable without relying on colour alone. The interface should also expose camera availability and processing errors so that a failed input stream is not mistaken for an alert-free state.

V. Implementation Stack

Table I summarises the technologies used or specified for the prototype. The final submitted report should retain only components that are present in the actual implementation.

TABLE II
REPORTED DRIVER-STATE CLASSIFICATION RESULTS

Metric	Reported value
Accuracy	92.4%
Precision	91.8%
Recall	93.1%
F1-score	92.4%
Test set size	More than 2,000 frames

VI. EXPERIMENTAL EVALUATION

The project evaluation reported the following metrics for the driver-state classification task. The values are included as project-reported results and should be retained only if they can be reproduced from the final test labels and prediction outputs.

Accuracy measures the fraction of correct predictions. Precision indicates how many predictions labelled Drowsy were correct, while recall indicates how many labelled Drowsy examples were detected. The F1-score is the harmonic mean of precision and recall. These metrics should be calculated from the same held-out test set and the positive class must be stated explicitly. If frames from the same video or subject appear in both training and test data, the reported values may overestimate generalisation.

A complete evaluation should include the confusion matrix, class-wise precision and recall, test-set class distribution, threshold and consecutive-frame settings, camera frame rate, and performance across illumination and head-pose conditions. Reporting only aggregate accuracy is insufficient to assess false alarms and missed drowsiness events.

VII. DISCUSSION

The EAR-based approach is lightweight and interpretable, making it suitable for a real-time prototype. The consecutive-frame rule helps prevent short blinks from immediately triggering a warning. However, the approach is sensitive to landmark quality and camera placement. Sunglasses, low light, strong head rotation, and face occlusion can lead to unreliable eye measurements. A fixed threshold may also behave differently across drivers.

The dual-camera architecture can provide complementary information: the inward-facing stream observes driver behaviour, while the forward-facing stream provides road context. However, using two cameras increases system complexity and compute requirements. The streams should be timestamped, monitored for failures, and tested independently. Object and lane detections should not be presented as safety guarantees, and a prototype warning system should not replace safe driving practices or certified driver-assistance equipment.

VIII. LIMITATIONS AND FUTURE WORK

The following improvements would strengthen the system and its evaluation:

- Calibrate EAR thresholds per user or use a validated temporal model.
- Evaluate on subject-independent and video-independent test splits.
- Report a confusion matrix and false-positive/false-negative rates.
- Test day, night, low-light, glasses, head-pose, and partial-occlusion scenarios.
- Benchmark end-to-end latency, frame rate, and compute use on the target device.
- Evaluate YOLOv8 road-object detection on a labelled road-scene test set.
- Implement and independently validate lane detection with appropriate lane-specific metrics.
- Add robust camera-failure handling and synchronised timestamps for both streams.

IX. CONCLUSION

This paper presented DriveSafe, a camera-based driver monitoring prototype that uses facial landmarks and the Eye Aspect Ratio to detect prolonged eye closure and generate a drowsiness warning. The project-reported evaluation achieved 92.4% accuracy, 91.8% precision, 93.1% recall, and 92.4% F1-score on more than 2,000 test frames for driver-state classification. The system design also supports a dual-camera configuration for driver monitoring and forward road-scene analysis. Further validation, including subject-independent testing and separate evaluation of object and lane detection, is necessary before making claims about deployment-level reliability or road-safety impact.

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